
Terraform Outputs

1 What Are Outputs in Terraform?

Outputs in Terraform are used to display useful information after the infrastructure is created.

Outputs allow you to:

- Print resource IDs to the terminal after `terraform apply`
- Pass information to other Terraform modules
- Verify that resources were created correctly
- Collect important values like VPC ID, Subnet IDs, NAT Gateway ID, etc.
- Help developers or students quickly view important infrastructure details

Outputs are especially important when:

- ✓ You need to connect resources across modules
- ✓ You need to share resource IDs with other tools
- ✓ You want Terraform to show final results clearly

2 Why Do We Use Outputs?

- ✓ Easily view important resource attributes

Example: After VPC creation, you can instantly see:

- VPC ID
- Subnet IDs
- NAT Gateway ID

✓ Useful for automation

Other modules or CI/CD pipelines might **consume the values**.

✓ Avoid manually searching in AWS Console

Outputs save time by showing everything upfront.

✓ Better visibility for students

After every apply, they see:

```
vpc_id = vpc-12345678  
public_subnet_id = subnet-abc12345
```

3 How Outputs Work in Terraform

Outputs have 3 parts:

★ 1. Output Name

A readable identifier:

```
output "vpc_id" {
```

★ 2. The value to output

Usually from a resource:

```
value = aws_vpc.main.id
```

★ 3. (Optional) Description

Helps students understand what the value is:

```
description = "ID of the VPC"
```

Example: Output Block Syntax

```
output "name" {  
  value      = <expression>  
  description = "What this output represents"  
}
```

4 Outputs Used in Your VPC Project

Here are the exact outputs you should create in [outputs.tf](#).

◆ 1. VPC ID

This helps you identify the VPC in AWS.

```
output "vpc_id" {
```

```
value    = aws_vpc.main.id
description = "The ID of the created VPC"
}
```

◆ 2. Public Subnet ID

Useful for launching EC2 instances or creating ALBs.

```
output "public_subnet_id" {
  value    = aws_subnet.public.id
  description = "The ID of the public subnet"
}
```

◆ 3. Private Subnet ID

Useful for RDS, private EC2, ECS tasks, etc.

```
output "private_subnet_id" {
  value    = aws_subnet.private.id
  description = "The ID of the private subnet"
}
```

◆ 4. NAT Gateway ID

Useful for debugging or referencing NAT Gateway in future modules.

```
output "nat_gateway_id" {
  value    = aws_nat_gateway.nat.id
  description = "The ID of the NAT Gateway"
}
```

★ Full `outputs.tf` File

Here is the complete version ready to paste:

```
output "vpc_id" {
  value     = aws_vpc.main.id
  description = "The ID of the created VPC"
}

output "public_subnet_id" {
  value     = aws_subnet.public.id
  description = "The ID of the public subnet"
}

output "private_subnet_id" {
  value     = aws_subnet.private.id
  description = "The ID of the private subnet"
}

output "nat_gateway_id" {
  value     = aws_nat_gateway.nat.id
  description = "The ID of the NAT Gateway"
}
```

5 How to View Outputs After Deployment

After `terraform apply`, Terraform prints:

Outputs:

```
vpc_id = "vpc-02b54d88f8b1c1234"  
public_subnet_id = "subnet-01a234bc56d23ef45"  
private_subnet_id = "subnet-0c45d1234bfed6789"  
nat_gateway_id = "nat-05ab12cd34ef56gh7"
```

6 How to Query Outputs Manually

You can re-display outputs anytime using:

```
terraform output
```

To show a specific one:

```
terraform output vpc_id
```

Conclusion

Terraform outputs turn your deployment from “*black box*” into “*transparent, visible infrastructure*.”

They provide:

- Visibility
- Reusability
- Integration capability
- Better debugging

- Automatic documentation

They are essential for building **professional, production-level Terraform projects**.
